

Project Topic

Title: Automated Text Summarization System Using Advanced Natural Language Processing (NLP)

Abstract:

The massive influx of digital text—ranging from lengthy research documents to extensive news feeds—demands efficient automated tools to extract critical information quickly. This project focuses on the development of an Automated Text Summarization System leveraging state-of-the-art Natural Language Processing (NLP) techniques. The system explores two primary methodologies: Extractive Summarization, which ranks and extracts key structural sentences using statistical algorithms like TF-IDF, and Abstractive Summarization, which utilizes deep learning Transformer models (such as BART or T5) to comprehend context and rewrite original, human-like summaries. The final outcome is a scalable NLP application capable of compressing long-form multi-page text documents into concise, grammatically correct bulleted points or short paragraphs while retaining 100% of the core semantic meaning.